Cost-Effective Solar-Powered Industrial Automation System with Priority-Based Load Control and Fault Protection

***Abstract*—This paper proposes a cost-effective ESP32-based solar-powered industrial automation system with integrated real-time monitoring, fault detection and priority-based load control. The proposed system combines photovoltaic energy generation, battery management, environmental sensing and intelligent automation within a unified embedded framework. It is designed to ensure efficient, reliable and sustainable industrial operation under dynamic conditions. Key parameters, including voltage, current, temperature, humidity and light intensity, are continuously monitored through integrated sensors. Edge-based processing enables rapid decision-making with low latency. The proposed system introduces fault isolation and priority-driven load management. Faulty or lower-priority loads are selectively disconnected, while critical machines remain operational. This approach improves system reliability and operational continuity. Simulation results demonstrate stable performance, maintaining approximately 5.1–5.2 V bus voltage, 1.0 A current, 5.0–5.3 W power output and up to 100% battery state of charge. The system effectively responds to predefined fault conditions, including overcurrent, undervoltage (below 4.5 V), overtemperature (above 40-45°C) and high humidity (above 75-80%). The proposed framework provides a scalable, energy-efficient and practical solution for small-scale industrial automation and off-grid applications. The system achieves rapid fault response with selective isolation and maintains stable operation under dynamic environmental conditions.**

***Index Terms*—ESP32, Fault Detection, Industrial Automation, IoT, Priority-Based Control, Solar Energy.**

I. Introduction

The rapid growth in global energy demand and environmental concerns has accelerated the adoption of renewable energy systems, particularly photovoltaic (PV) technologies, due to their scalability and cost efficiency [1]. However, the intermittent nature of solar energy introduces challenges in maintaining system stability, efficient energy utilization and real-time control [2]. These challenges become more critical in industrial environments, where continuous operation and reliability are essential for maintaining productivity and minimizing operational losses.

The integration of the Internet of Things (IoT) enables real-time monitoring and control of industrial systems through continuous acquisition of electrical and environmental parameters [3]. By leveraging embedded sensors and communication technologies, IoT-based systems can improve visibility and support data-driven decision-making. Additionally, recent works incorporate data-driven techniques for improved fault detection and predictive maintenance [4]. These approaches enhance system reliability by identifying anomalies at an early stage and reducing downtime.

Despite these advancements, many existing systems still lack unified integration of monitoring, control and automation functionalities [5]. In most cases, energy management, fault detection and load control are treated as separate modules, leading to increased system complexity and inefficiency. Furthermore, reliance on cloud-based processing introduces latency and limits real-time responsiveness in industrial environments [6]. This dependency also raises concerns regarding network reliability and scalability, especially in remote or off-grid applications.

Another critical limitation in conventional systems is the lack of intelligent load prioritization. During energy shortages or fault conditions, many systems perform complete shutdowns, which can interrupt critical operations and reduce overall system efficiency. This highlights the need for an integrated solution capable of adaptive control and selective operation based on system conditions.

To address these limitations, this paper proposes an ESP32-based embedded architecture that integrates solar energy management, real-time monitoring, fault detection and priority-based load control within a single platform. The proposed system employs machine-level fault isolation and priority-driven automation, allowing only faulty or low-priority loads to be disconnected while critical machines remain operational. This approach enhances system reliability, improves energy utilization and ensures continuous operation under dynamic and constrained energy conditions.

II. Literature Review

The integration of renewable energy systems with IoT-based monitoring has been widely explored to improve efficiency, reliability and fault tolerance. IoT-enabled smart grid monitoring systems support real-time data acquisition and control for distributed energy resources [7]. Early PV monitoring systems demonstrated improved maintenance, remote visualization and system diagnostics through cloud-based platforms [8]. These systems provide enhanced accessibility but often depend on continuous network connectivity, which may limit their applicability in remote or off-grid environments.

Solar energy optimization has been further enhanced through dual-axis tracking systems using ESP32 and LDR sensors [9], as well as advanced tracking algorithms for improved energy harvesting efficiency [10]. These approaches significantly increase solar output but typically focus only on energy optimization without integrating load control or fault management. Industrial IoT monitoring frameworks have also been developed to improve equipment reliability and fault detection [11]. Similarly, low-cost PV monitoring systems using ESP32 have demonstrated promising results in small-scale

deployments [12], highlighting the feasibility of embedded-based solutions.

Recent approaches incorporate machine learning for anomaly detection and fault classification in PV systems [13], [14]. These methods improve detection accuracy and predictive maintenance capabilities but often require high computational resources and cloud-based processing, increasing system complexity and latency. Hybrid renewable monitoring systems integrating solar and wind energy have also been explored [15], [16], offering improved energy availability but introducing additional system design challenges and cost considerations.

TABLE I: Comparison of Existing Systems with Proposed Approach

Reference	IoT	Solar	Fault Detection	Priority Control	Edge-Based
[7]	✓	✓	×	×	×
[8]	✓	✓	×	×	×
[9]	✓	✓	×	×	×
[10]	×	✓	×	×	×
[11]	✓	×	✓	×	×
[12]	✓	✓	×	×	×
[13]	✓	✓	✓	×	×
[14]	✓	✓	✓	×	×
[15]	✓	✓	×	×	×
[16]	✓	✓	×	×	×
Proposed System	✓	✓	✓	✓	✓

Table I compares key functional aspects including IoT integration, solar utilization, fault detection capability, priority-based control and edge processing highlighting that existing systems primarily focus on monitoring, tracking or fault detection independently, whereas the proposed system integrates all functionalities along with priority-based control and edge processing.

Despite these advancements, most existing systems treat monitoring, tracking and control as independent modules, resulting in fragmented architectures and inefficient system coordination. Furthermore, limited attention has been given to intelligent load prioritization under constrained energy conditions, where maintaining operation of critical loads is essential. In contrast, this work proposes a unified, low-cost embedded solution that integrates real-time monitoring, solar energy management, fault detection and priority-based load control within a single platform. The use of edge-based processing further reduces latency and enhances real-time responsiveness, making the system suitable for practical industrial and off-grid applications.

III. System Architecture

A. Overall Design

The proposed system is an ESP32-based IoT-enabled industrial automation platform powered by solar energy. It integrates real-time monitoring, fault detection and priority-based load control within a unified embedded architecture. The system is structured into sensing, processing, power management and actuation layers to ensure efficient and modular operation.

Fig. 1 shows the integration of solar power generation, sensing, ESP32-based control and priority-driven automation of industrial loads.

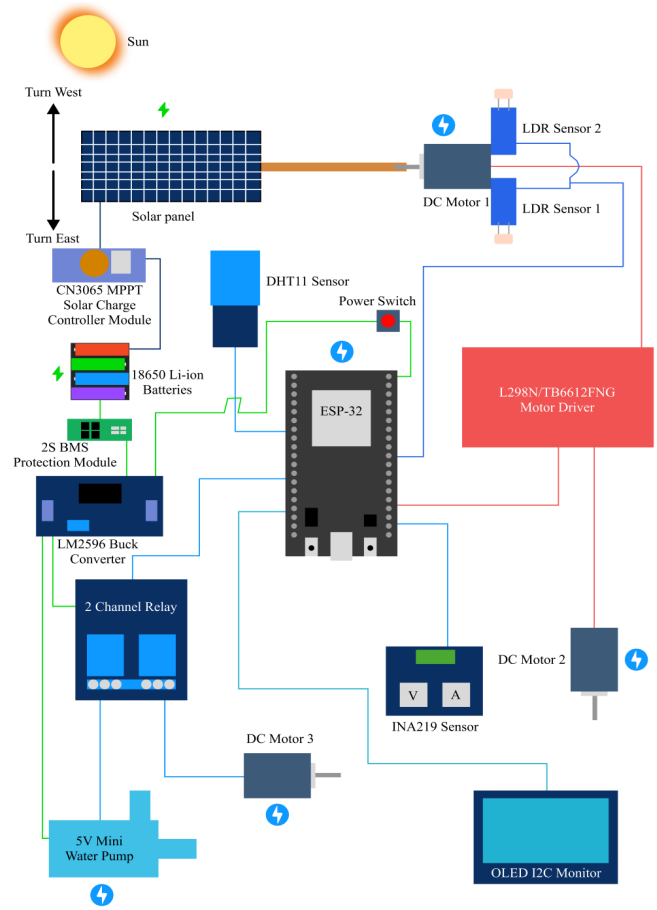

Fig. 1: Proposed system architecture.

B. Hardware Architecture

The ESP32 acts as the central controller, acquiring electrical data using the INA219 sensor and environmental data using the DHT11 sensor. An LDR pair enables solar tracking.

The power subsystem consists of a solar panel, CN3065 charge controller, 18650 battery pack, BMS protection module and LM2596 buck converter, ensuring stable energy generation and regulated supply.

Actuation is handled using a motor driver and relay module controlling three DC motors and a water pump, while an OLED display provides real-time system feedback.

C. Control Flow

The ESP32 continuously processes sensor data and applies threshold-based fault detection. Faulty or low-priority loads are selectively disconnected instead of shutting down the entire system. This enables low-latency operation and efficient energy utilization under dynamic conditions.

D. Cost Analysis

The proposed system is designed to be low-cost and suitable for practical deployment. The total cost of the prototype is estimated based on local market prices in Bangladesh, as shown in Table II.

TABLE II: Cost Analysis of the Proposed System (USD)

Component	Quantity	Price (USD)
ESP32 Dev Module	1	\$8.00
Solar Panel (6V, $\geq$ 3W)	1	\$10.00
CN3065 Charge Controller	1	\$5.00
18650 Batteries	2	2 $\times$ \$5.00 = \$10.00
BMS Protection Module	1	\$3.00
LM2596 Buck Converter	1	\$3.00
LDR Sensors	2	2 $\times$ \$1.00 = \$2.00
INA219 Sensor	1	\$4.00
DHT11 Sensor	1	\$3.00
DC Motors	3	3 $\times$ \$5.00 = \$15.00
Water Pump	1	\$6.00
Motor Driver (TB6612FNG)	1	\$6.00
Relay Module	1	\$3.00
OLED Display	1	\$8.00
Miscellaneous Components	1	\$10.00
Total Cost		**\$96.00**

The cost estimation is based on average market prices in the United States and may vary depending on supplier and region.

The total cost indicates that the proposed system can be implemented at a relatively low cost, making it suitable for small-scale industrial and renewable energy applications.

This architecture ensures seamless integration of sensing, decision-making and actuation within a compact embedded framework.

IV. Methodology

A. Proposed Automation Method

The proposed system implements a priority-based industrial automation strategy using the ESP32 microcontroller. Multiple loads, including three DC motors and one water pump, are connected, each with predefined limits for voltage, current, temperature, humidity and load demand. Instead of shutting down the entire system under abnormal conditions, the controller evaluates each machine individually and disconnects only faulty or low-priority loads. This ensures the uninterrupted operation of critical machines under constrained energy conditions.

The automation logic is based on three key decisions: fault detection, priority-based load control and energy availability. When a machine exceeds its threshold, it is immediately isolated. If available energy becomes insufficient, low-priority machines are disabled first, allowing high-priority loads to continue operating.

B. Solar Energy and Battery Charging

The system is powered by a solar energy unit. The solar panel converts sunlight into electrical energy, which is regulated by the CN3065 charge controller to charge the 18650 battery pack. The BMS module provides protection against overcharging, over-discharging and unsafe current conditions. The LM2596 buck converter ensures stable voltage supply to the ESP32 and connected components.

For improved energy harvesting, LDR sensors are used for solar tracking. The ESP32 compares light intensity from both sensors and adjusts panel orientation accordingly, maximizing charging efficiency.

C. Mathematical Decision Logic

The instantaneous power is calculated as:

$$P(t) = V(t) \times I(t) \tag{1}$$

This equation computes the instantaneous electrical power based on the measured voltage and current values, where $P(t)$ is the power, $V(t)$ is the voltage and $I(t)$ is the current at time t.

Battery condition is evaluated using:

$$SOC < SOC_{min} \tag{2}$$

This condition evaluates whether the battery state of charge falls below the minimum threshold required for stable operation, where SOC is the current battery state of charge and SOC_{min} is the minimum allowable charge level.

A machine is considered faulty if:

$$F_i(t) = \begin{cases} 1, & \begin{array}{l} T_i > T_{max,i} \\ \text{or } H_i > H_{max,i} \\ \text{or } I_i > I_{max,i} \end{array} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

This equation defines the fault condition of a machine based on threshold violations, where $F_i(t)$ represents the fault state of machine i, T_i, H_i and I_i denote temperature, humidity and current and $T_{max,i}$, $H_{max,i}$ and $I_{max,i}$ are their respective maximum limits.

Priority-based control is defined as:

$$L_i = \begin{cases} 0, & \text{if low priority and } SOC < SOC_{min} \\ 1, & \text{otherwise} \end{cases} \tag{4}$$

This condition determines whether a machine remains active based on priority and energy availability, where L_i represents the operational state of machine i (0 = off, 1 = active).

Solar tracking is determined by:

$$\Delta L = L_{left} - L_{right} \tag{5}$$

where ΔL defines the direction of panel movement.

This equation calculates the light intensity difference to determine the direction of panel movement, where L_{left} and L_{right} represent the light intensity measured by the left and right LDR sensors, respectively.

These decision models allow the system to perform real-time evaluation of both electrical and environmental conditions. By combining threshold-based fault detection with priority-driven load control, the system ensures efficient operation under dynamic scenarios. The integration of these mechanisms enables rapid response to faults while maintaining system stability and minimizing unnecessary shutdowns.

D. Workflow of the Proposed System

Fig. 2 illustrates the complete operational workflow of the proposed system. Initially, the ESP32 collects real-time data from electrical sensors (INA219), environmental sensors (DHT11) and light sensors (LDR). The collected data is processed to evaluate system conditions, including voltage

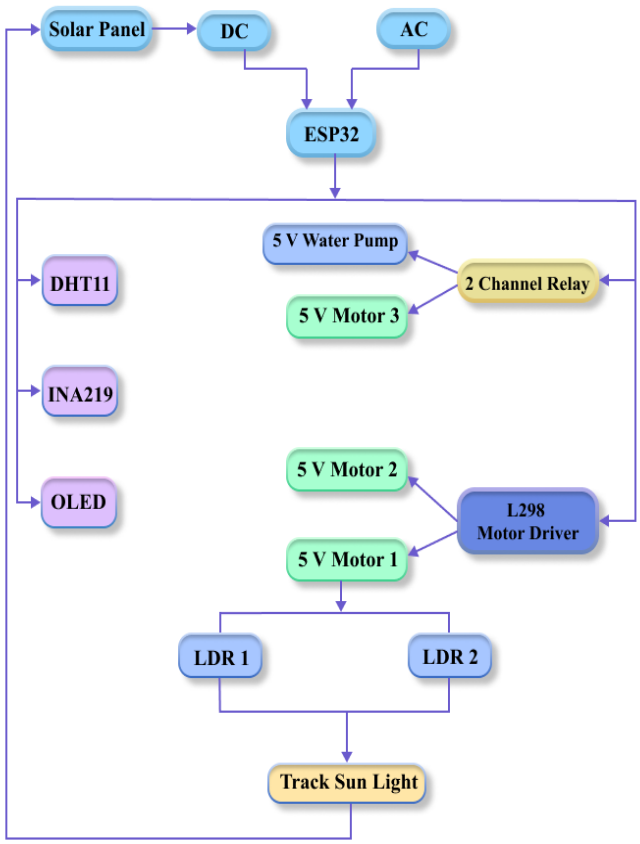

Fig. 2: Workflow of the proposed ESP32-based priority automation and fault protection system.

stability, current levels, temperature, humidity and battery state of charge.

The system then performs machine-level fault analysis by comparing sensor values against predefined thresholds. If a fault condition is detected, the corresponding machine is immediately isolated. In parallel, the system evaluates energy availability and load demand to determine whether priority-based control is required.

Based on these decisions, the system either maintains normal operation, selectively shuts down faulty machines or disables low-priority loads to support critical operations. The final system status is displayed on the OLED and logged through the serial interface.

E. Work in Process

The working process of the proposed system is described step by step below:

1) Solar energy generation and battery charging.
2) Sensor data acquisition (INA219, DHT11, LDR).
3) Real-time threshold evaluation.
4) Fault detection and selective shutdown.
5) Priority-based load control.
6) OLED display and serial logging.

F. C++ Implementation: Pseudocode

The automation logic is implemented using a structured control algorithm, as presented below:

Algorithm 1 Priority-Based Industrial Automation

```
Start system
Initialize ESP32, sensors, OLED, relays and motor driver
while system is ON do
    Read voltage, current, temperature, humidity, battery level and LDR values
    Display real-time values on OLED
    for each connected machine do
        if machine threshold is exceeded then
            Shut down only that machine
            Log fault condition
        end if
    end for
    if battery level is low or load demand is high then
        Stop low-priority machines first
        Keep high-priority machines running
    end if
    if left LDR and right LDR values are different then
        Rotate solar panel toward higher light intensity
    else
        Keep solar panel stable
    end if
end while
End
```

V. Simulated Results and Discussion

The simulation was developed to validate the proposed ESP32-based industrial automation system under normal and fault conditions. The simulator integrates solar input control, battery charging, INA219-based electrical monitoring, DHT11-based environmental sensing, LDR-based solar tracking, OLED display, serial logging, manual control and per-device fault protection for three DC motors and one water pump.

Under normal operation, all four loads remain active with stable system parameters. The system maintains approximately 5.1-5.2 V bus voltage, 1.0 A current and 5.0-5.3 W power, along with 30°C temperature, 55% humidity and up to 100% battery state of charge, confirming proper energy generation and regulation.

The simulator includes predefined automated fault scenarios, namely overcurrent fault, undervoltage trip, overheat shutdown, high humidity trip, overload demand and normal operation. These scenarios dynamically modify system conditions and trigger protection mechanisms.

In fault conditions, the system performs a selective machine-level shutdown. Overcurrent and undervoltage conditions result in immediate disconnection of affected loads to prevent system instability. During overtemperature conditions (e.g., 45°C), machines exceeding their limits (Motor 2 and water pump) are automatically shut down, while other machines continue operating. Similarly, in high humidity conditions

(e.g., 82%), only machines exceeding their thresholds are disconnected.

The overload demand scenario demonstrates priority-based automation, where low-priority machines are turned off first to ensure uninterrupted operation of critical loads. This confirms that the system avoids a complete shutdown and instead optimizes energy distribution.

A. Simulated Results

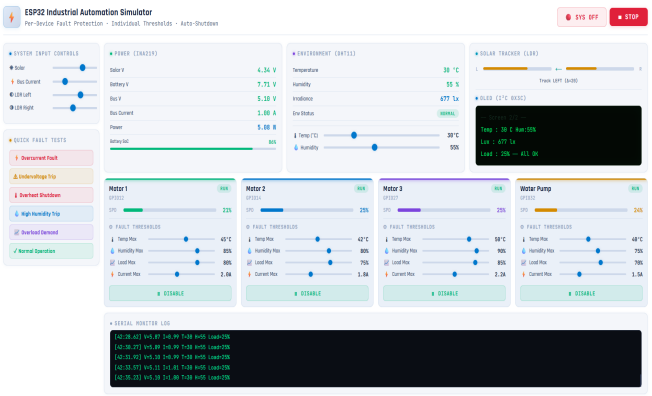

Fig. 3: Normal operating condition where all machines remain active.

Fig. 3 shows stable system operation where all machines remain active under normal conditions with balanced power distribution and no fault triggers.

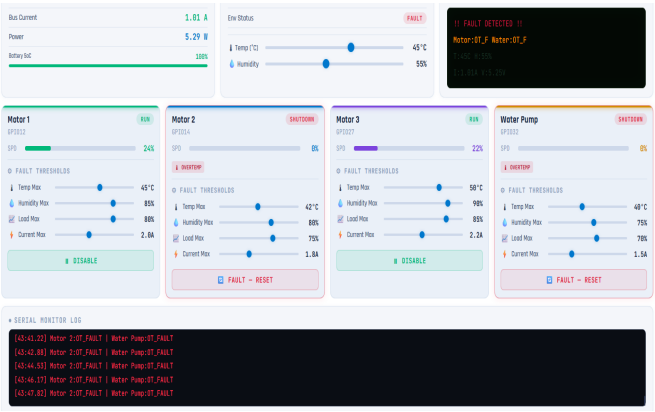

Fig. 4: Selective shutdown during overtemperature condition.

Fig. 4 illustrates selective fault isolation, where only machines exceeding temperature thresholds are shut down while others continue operating.

Fig. 5 demonstrates priority-based control under high humidity, where only affected machines are disconnected without interrupting the entire system.

The solar charging subsystem operates automatically, where solar energy is continuously stored in the battery through the charge controller and BMS protection. The system dynamically adjusts power distribution based on battery state of charge and load demand.

Fig. 5: Priority-based shutdown during high-humidity condition.

The LDR-based solar tracking mechanism also functions correctly by adjusting panel orientation based on light intensity differences. Overall, the simulation confirms effective automation, reliable fault handling and efficient energy-aware control of industrial loads.

B. Performance Analysis

The performance of the proposed system was evaluated under multiple operating conditions to assess its reliability and adaptability. Under normal conditions, the system maintained stable electrical parameters, including 5.1-5.2 V bus voltage, 1.0 A current and 5.0-5.3 W power output. This confirms efficient energy regulation and consistent operation.

During fault scenarios, the system demonstrated rapid response through selective machine-level isolation. Overcurrent and undervoltage conditions resulted in immediate disconnection of affected loads, preventing system instability. In overtemperature and high humidity scenarios, only machines exceeding predefined thresholds were shut down, while other machines continued operating.

The priority-based control mechanism further improved system performance by ensuring uninterrupted operation of critical loads during limited energy availability. Compared to conventional full-shutdown systems, the proposed approach reduces downtime and enhances operational continuity.

Overall, the system demonstrates stable performance, effective fault handling and efficient energy utilization, validating its suitability for real-time industrial automation.

C. Key Advantages of the Proposed System

The proposed system offers several advantages over existing industrial automation solutions:

- **Selective Fault Isolation:** Only faulty machines are disconnected, avoiding complete system shutdown.
- **Priority-Based Load Control:** Ensures uninterrupted operation of critical loads under limited energy conditions.
- **Energy Efficiency:** Solar energy integration and intelligent distribution improve overall system efficiency.

- **Low-Cost Implementation:** The system can be developed at approximately $96, making it accessible for small-scale applications.
- **Edge-Based Processing:** Eliminates dependency on cloud systems, reducing latency and improving real-time response.

D. System Performance Evaluation

The proposed system was further evaluated in terms of operational stability, fault response and energy efficiency. Under normal conditions, the system maintained consistent electrical parameters, demonstrating stable voltage regulation and load handling capability.

The fault response performance confirms that the system reacts immediately to abnormal conditions. In overcurrent and undervoltage scenarios, affected loads are disconnected without delay, preventing system-wide instability. Similarly, in environmental fault conditions such as high temperature and humidity, only machines exceeding thresholds are isolated, ensuring minimal disruption.

The priority-based control mechanism significantly enhances system efficiency. Instead of shutting down all loads, the system intelligently reduces load demand by disabling low-priority machines first. This improves energy utilization and ensures continuous operation of critical loads even under limited power conditions.

Additionally, the integration of solar energy and battery storage enables uninterrupted operation. The system dynamically balances energy generation and consumption, maintaining stable performance even under fluctuating environmental conditions.

Overall, the results validate that the proposed system achieves reliable automation, efficient energy management and robust fault handling. These results demonstrate that the proposed system effectively balances performance, reliability and energy efficiency under diverse operating conditions.

E. Availability of Simulation Model

The simulation model developed in this study is publicly available for reproducibility and further research. It includes the ESP32-based control logic, fault scenarios and priority-based automation framework used in this work. The repository can be accessed at:

https://github.com/RagibNadim/Industrial-Monitoring-System-ESP32.git.

The repository includes test scenarios such as overcurrent, undervoltage, overtemperature, high humidity and overload demand.

VI. Conclusion

This paper presented an ESP32-based solar-powered industrial automation system integrating real-time monitoring, fault detection and priority-based load control. The system enables efficient energy management using solar charging, battery storage and edge-based processing. The proposed system demonstrates a strong balance between cost, performance and reliability. Its modular design allows easy scalability for larger industrial applications. Furthermore, the use of edge-based processing reduces latency and enhances real-time responsiveness compared to cloud-dependent systems.

The main contribution is the implementation of machine-level fault isolation and priority-driven automation, where only faulty or low-priority loads are disconnected while critical machines continue operating. This improves system reliability and ensures better utilization of limited energy resources. Simulation results demonstrate that the proposed system successfully performs automated control under varying conditions, maintaining stable operation and reducing unnecessary shutdowns. The low-cost design further supports its suitability for small-scale industrial and off-grid applications.

Additionally, the system demonstrates strong adaptability under varying environmental and load conditions. The combination of renewable energy integration and intelligent control makes it a promising solution for sustainable industrial automation. Future work will include real-time hardware validation and quantitative evaluation of system latency, response time and fault detection accuracy with integration of predictive intelligence for enhanced system performance.

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